

The Mean Value Theorem

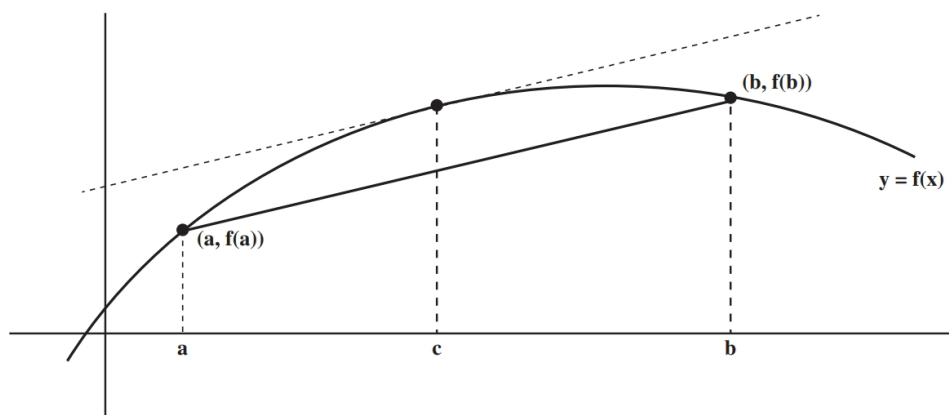
Section 4.2

A. Rolle's Theorem

Let $f(x)$ be a function such that $f(x)$ is continuous on $[a, b]$, $f(x)$ is differentiable on (a, b) and $f(a) = f(b)$, then there is a number c in (a, b) such that $f'(c) = 0$

B. The Mean Value Theorem

Let $f(x)$ be a function such that $f(x)$ is continuous on $[a, b]$ and $f(x)$ is differentiable on (a, b) , then there is a number c in (a, b) such that $f'(c) = \frac{f(b) - f(a)}{b - a}$, or equivalently $f(b) - f(a) = f'(c) \cdot (b - a)$



C. Constant Theorem

If $f'(x) = 0$ for all x in an interval (a, b) then $f(x)$ is constant on (a, b) .

D. Corollary

If $f'(x) = g'(x)$ for all x in an interval (a, b) then $f - g$ is constant on (a, b)
(i.e., $f(x) = g(x) + c$)

Examples

1. **(video)** Consider the function $f(x) = 3x^2 - 6x + 4$ on the interval $[-1, 3]$. Verify that this function satisfies the three hypotheses of Rolle's Theorem on the interval.

$f(x)$ is _____ on $[-1, 3]$

$f(x)$ is _____ on $[-1, 3]$

and $f(-1) = f(3) =$ _____

Then by Rolle's theorem, there exists a c such that $f'(c) = 0$. Find the value c .

2. **(video)** Consider the function $f(x) = 3x^2 + x + 4$ on the interval $[-1, 4]$. Find the average or mean slope of the function on this interval, that is,

$$\frac{f(4) - f(-1)}{4 - (-1)} =$$

By the Mean Value Theorem, we know there exists a c in the open interval $(-1, 4)$ such that $f'(c)$ is equal to this mean slope. For this problem, there is only one c that works.

$c =$ _____

3. Consider the function $f(x) = x^2 - 4x + 1$ on the interval $[0, 4]$. Verify that this function satisfies the three hypotheses of Rolle's Theorem on the interval.

$f(x)$ is _____ on $[0, 4]$; $f(x)$ is _____ on $(0, 4)$; and $f(0) = f(4) =$ _____

Then by Rolle's Theorem, there exists a number c such that $f'(c) = 0$. Find the value c .

4. Consider the function $f(x) = 3x^2 + 5x + 11$ on the interval $[-3, 6]$. Find the average or mean slope of the function on this interval, that is, $\frac{f(6) - f(-3)}{6 - (-3)}$

By the Mean Value Theorem, we know there exists a c in the open interval $(-3, 6)$ such that $f'(c)$ is equal to this mean slope. For this problem, there is only one c that works. $c =$ _____

5. By applying Rolle's Theorem, check whether it is possible that the function $f(x) = x^5 + x - 11$ has two real roots.

Possible or impossible?

Your reason is that if $f(x)$ has two real roots then by Rolle's Theorem: $f'(x)$ must be

_____ at certain value of x between these two roots, but $f'(x)$ is always negative, positive, or zero _____

6. Suppose $f(x)$ is continuous on $[2, 8]$ and $3 \leq f'(x) \leq 9$ for all x in $(2, 8)$. Use the Mean Value Theorem to estimate $f(8) - f(2)$.

_____ $\leq f(8) - f(2) \leq$ _____